

The Capacity Cliff Is Not Capacity: Locating, Dissolving, and Exonerating a Coherence Confound in Trained Fast-Weight Memories

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Abstract

A trained DeltaNet-style fast-weight memory that recruits provably-necessary rank to solve an associative-recall task does not degrade gracefully as load approaches capacity: it falls off a cliff. We locate that cliff precisely — at capacity ratio $K/d_{\text{state}} = 0.5455$ (95% seed-level bootstrap CI $[0.5385, 0.5513]$, width 0.0127, $d_{\text{state}} = 64$) — by measuring exact held-out compositional recovery across four new intermediate loads ($K = 34, 38, 42, 46$) bracketed by two previously-archived anchors ($K = 32, 48$). We then ask whether this is a universal property of the ratio K/d_{state} or a finite-size artifact of $d_{\text{state}} = 64$ specifically, by re-running the identical K/d_{state} window at $d_{\text{state}} = 128$ ($K = 68, 76, 84, 92$): the cliff is gone, $h4 = 1.0$ at all four loads, all seeds — not merely shifted, but absent from the entire measured window. The natural mechanistic candidate is anchor-table *coherence* ($\max |\cos|$ among the table’s rows, forced upward when the entity count exceeds d_{state} by the Welch bound, and removed entirely when it does not): we test this directly by injecting controlled, frozen coherence into an otherwise-flat ($d_{\text{state}} = 128$) table and sweeping the dose from 0 up to 0.40 — exceeding $d_{\text{state}} = 64$ ’s own measured trained-coherence band (0.373–0.385) — under a concentrated (rank-4) injection structure. The result is a clean exoneration: $h4 = 1.0$ at every dose, 10/10 cells, no exception. Scalar coherence, injected directly at and above the level the real cliff co-occurs with, does not reproduce it. Two candidates survive, neither adjudicated by data in hand: overlap *structure* (a diffuse rather than concentrated injection may behave differently) and raw *state capacity* (d_{state} grew $4\times$ while K only grew $\sim 2\times$ across the two waves, a confound this design cannot separate). We report a precisely-located transition, its clean dissolution with scale, and the honest failure of the most natural single-variable account of it — a complete, three-act, negative-result-driven capacity story with every reported number traced to an archived run.

1 Introduction

The rank-recruitment question. Descriptive work on linear-attention and fast-weight LLMs shows that the state’s effective rank is bounded by sequence length and by how many independent associations the state must hold [4, 9]. A companion causal result on this same DeltaNet testbed [7] showed something stronger: when a task *forces* the state to solve K independent associative bindings under a hard single-state bottleneck, gradient descent recruits effective rank $\approx K$ and the resulting write operator is exact, not merely approximately correct, for a wide range of K . That raises the natural next question this paper answers: **what happens as K approaches the state’s own dimensionality d_{state} ?** A linear memory has a hard information-theoretic ceiling near $K \approx d_{\text{state}}$; but a *trained* memory need not approach that ceiling gracefully, and prior work in this testbed lineage already hinted it does not [1]. This paper measures exactly how the trained system fails as load increases, locates the failure precisely, tests whether it is fundamental to the architecture or an artifact of one scale, and then tests the single most natural mechanistic account of it directly.

Contribution 1 — the cliff, located. At $d_{\text{state}} = 64$, held-out compositional recovery ($h4 = \text{rec}@0.9$ at hop depth 4) does not decline smoothly with K/d_{state} — it

transitions sharply. We fit a logistic to six points spanning $K/d_{\text{state}} \in [0.25, 0.75]$ and locate the transition’s midpoint to $x_0 = 0.5455$ (95% CI $[0.5385, 0.5513]$, width 0.0127, characteristic width $w = 0.0597$), a $\sim 20\times$ tightening over the prior 0.25-wide bracket between the two flanking anchors.

Contribution 2 — the cliff, dissolved. Re-running the identical K/d_{state} window at $d_{\text{state}} = 128$ shows no cliff at all: $h4 = 1.0$ at all four matched loads, all seeds, 12/12 cells. The transition is not shifted — it exits the measured window entirely. This rules out a naive “ K/d_{state} ratio alone determines the cliff” account.

Contribution 3 — the leading confound, tested and exonerated. The two waves above differ in more than raw scale: the $d = 64$ table (107 entities packed into 64 dimensions) is forced non-orthogonal by the Welch bound, while the $d = 128$ table (107 entities, 128 dimensions) is exactly orthogonal. This is the single most natural remaining mechanistic candidate — and we test it directly, not by inference. Injecting controlled, frozen coherence into the otherwise-flat $d = 128/K = 68$ cell, swept from 0 to 0.40 (exceeding $d = 64$ ’s own trained coherence band), produces no cliff at any dose under a concentrated (rank-4) injection structure: $h4 = 1.0$, 10/10 cells, no exception.

What this paper is NOT. Not a claim that coherence-as-

a-scalar can never matter — only that this specific, concentrated (rank-4) injection, at this specific geometry, does not reproduce the cliff; a diffuse injection structure is a registered, not-yet-run follow-on (§4). Not a resolution of *why* the cliff occurs at $d = 64$ specifically — this paper locates and characterizes the transition and rules out one candidate mechanism; it does not close the mechanism question. Not a claim about production-scale Gated-DeltaNet [2, 8] or matrix-valued RNNs at scale [3] — all experiments here use a $\sim$ sub-1M parameter, single-layer synthetic-grammar testbed with 107 trainable entities.

2 Testbed and Process

2.1 Architecture and task

- Vanilla single-head, single-layer DeltaNet (chunk_delta_rule, production fla-org kernel), state update $S_t = S_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$ [7].
- Synthetic-grammar associative-recall task: a pool of 107 trainable entities, each bound to a value via the delta rule; recovery is graded by *exact continuous* cosine similarity against the true value at threshold 0.9 (rec@0.9), never argmax over a codebook — the design choice that keeps a rank bound from being silently satisfiable by codebook lookup [5].
- Held-out compositional recovery at hop depth 4, $h_4 = \text{rec}@0.9$ on entities/bindings never seen during training, is the paper’s single load-bearing quantity throughout.
- **The load parameter, K .** K is the number of independent entity bindings the state must simultaneously support within one episode; K/d_{state} is this paper’s capacity ratio.
- **The anchor table.** A per-entity key-anchoring mechanism (established in prior work on this testbed as improving held-out recovery through blend-arithmetic constancy, not learned entity-specific content) provides the frozen table this paper’s dose-response arm injects controlled coherence into.

2.2 Coherence, defined

- Anchor-table coherence is $\max |\cos|$ among the table’s $n = 107$ row vectors. The Welch bound forces this to be strictly positive whenever $n > d_{\text{state}}$ (as at $d = 64$, measured 0.2842 at construction) and permits exact orthogonality when $n \leq d_{\text{state}}$ (as at $d = 128$, measured 0.000000 at construction) — the single structural difference between the two scales beyond raw dimension.

2.3 Process: pre-registration and independent adversarial audit

- Every wave below was pre-registered (hypothesis and outcome semantics fixed before data existed) and independently attacked before being treated as final. The cliff-localization wave’s own deliverable (a CI width bar) was pre-registered against a power simulation before any of the four new K cells ran. The dimension-generalization wave’s three-way outcome space (CONFIRM-UNIVERSAL / CONFIRM-SHIFTED / degenerate) was fixed before any $d=128$ cell ran, including the explicit disclosure rule that a degenerate sigmoid fit on flat data must be reported as such, never quoted as a located transition.
- The dose-response wave’s own launch was gated on a pre-launch smoke test proving the frozen anchor table’s gradient path is truly severed (anchor_table.weight.grad is None) and that its coherence is bit-identical across the full 20,000-step run — verified after the fact at every one of 10 checkpoints per cell, zero drift, zero slack used.
- Instrument-saturation is checked, not assumed, at every wave: hop-21 recovery (a harder, far-extrapolation quantity than h_4) is never uniformly 1.0 in any of the three waves below, ruling out a globally-ceilinged read-out as an alternative explanation for any flat result reported in §3.

3 Results: The Capacity Trilogy

3.1 Act 1: the cliff, located

- At $d_{\text{state}} = 64$, six points spanning $K/d_{\text{state}} \in [0.25, 0.75]$ (Fig. 1, left) trace a sharp, not gradual, transition: $h_4 = 1.000$ ($K=16$) $\rightarrow$ 0.6669 ($K=32$) $\rightarrow$ 0.5676 ($K=34$) $\rightarrow$ 0.3316 ($K=38$) $\rightarrow$ 0.1177 ($K=42$) $\rightarrow$ 0.0434 ($K=46$) $\rightarrow$ 0.0215 ($K=48$).
- A logistic fit (0/4000 degenerate bootstrap fits) locates the transition’s midpoint at $x_0 = 0.5455$ (95% CI [0.5385, 0.5513], width 0.0127) with characteristic width $w = 0.0597$ (CI [0.0557, 0.0642]) — i.e. the transition’s effective span is $K/d_{\text{state}} \approx [0.486, 0.605]$, or $K \approx 31\text{--}39$ raw at $d=64$. This is a $\sim 20\times$ tightening of the prior 0.25-wide bracket (spanned only by the two flanking archived anchors) to a ± 0.006 -wide located transition.
- **This is a WHERE result, not a WHY result.** The fit characterizes the transition’s location and width; it makes no claim about the mechanism driving it (Acts 2–3 test the leading candidate directly).

The capacity cliff is dimension-dependent: located at $d_{\text{state}} = 64$, dissolved by $d_{\text{state}} = 128$ (same K/d window, candidate (d))

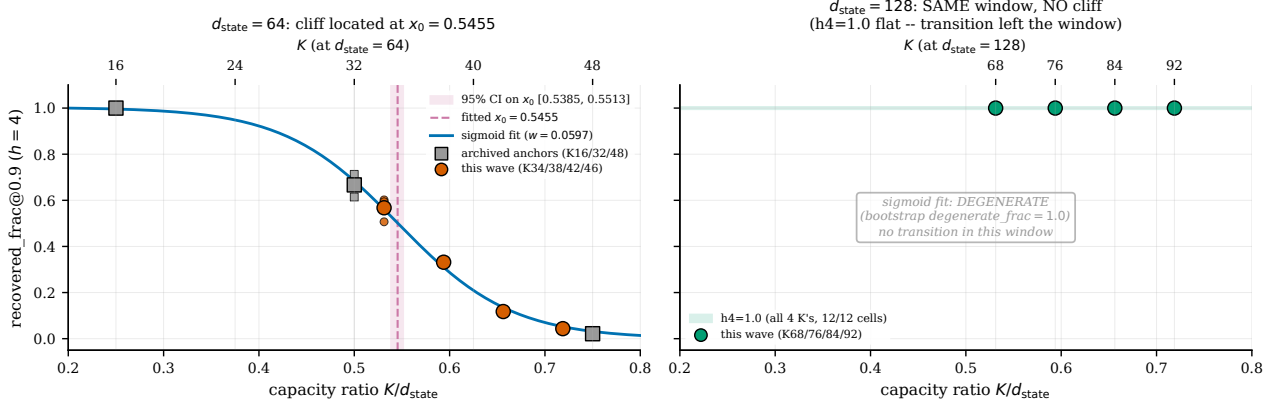


Figure 1: Left: at $d_{\text{state}}=64$, held-out h_4 recovery transitions sharply as K/d_{state} increases; a logistic fit to six points (two archived anchors, $K=32/48$, plus four new intermediate loads, $K=34/38/42/46$) locates the midpoint at $x_0 = 0.5455$ (95% CI [0.5385, 0.5513]). **Right:** re-running the identical K/d_{state} window at $d_{\text{state}}=128$ ($K=68/76/84/92$) shows no transition anywhere in the window — $h_4=1.0$ at all 12 cells. The sigmoid fit on this flat data is degenerate by construction (bootstrap degenerate fraction = 1.0) and is plotted as such, never as a spurious curve.

3.2 Act 2: the cliff, dissolved

- Re-running the identical K/d_{state} window at $d_{\text{state}} = 128$, using the equivalent loads $K = 68, 76, 84, 92$, shows $h_4 = 1.0$ at all four loads, all 3 seeds each, 12/12 cells, no exception (Fig. 1, right). This is the same K/d_{state} window that collapsed $0.667 \rightarrow 0.022$ at $d = 64$.
- Per pre-registered outcome semantics this is **CONFIRM-SHIFTED in its strong form**: not merely a shifted x_0 , but the transition exits the measured window entirely. At $d = 128$, capacity extends past $K/d_{\text{state}} = 0.71875$ — at least 92 of 107 entities recover perfectly through held-out 4-hop composition, in the exact band where $d = 64$ had already collapsed to near zero.
- Instrument-saturation is ruled out: hop-21 recovery at $d = 128$ ranges 0.9900–1.0 across the 12 cells (not uniformly 1.0), and effective rank tracks each K almost exactly (e.g. $K=84 \rightarrow 83.6$) — the state is using its full permitted rank, not saturating against d_{state} itself.
- **This rules out “ K/d_{state} ratio alone determines the cliff.”** The exact ratio window that cliffs at $d = 64$ stays completely flat at $d = 128$.

3.3 Act 3: the leading confound, tested and exonerated

- The two waves above differ in more than raw scale: the $d = 64$ table (107 entities in 64 dimensions) is forced non-orthogonal by the Welch bound (measured

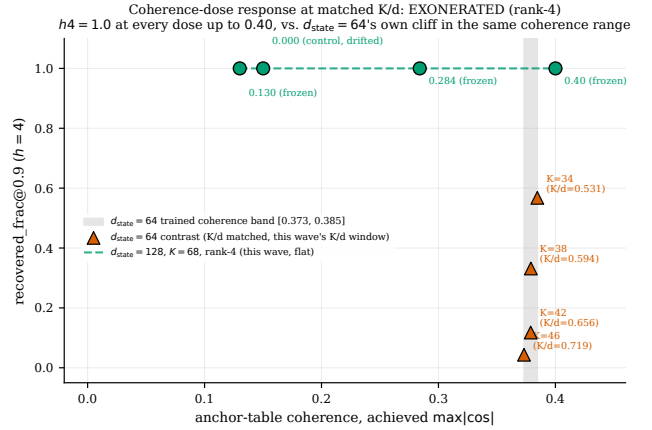


Figure 2: Coherence-dose response at $K=68$, $d_{\text{state}}=128$, rank-4 (concentrated) injection: h_4 stays flat at 1.0 at every dose tested (green), including 0.40 — exceeding $d_{\text{state}}=64$'s own measured trained-coherence band (grey, [0.373, 0.385]) — while $d_{\text{state}}=64$'s own cliff (orange) collapses across the identical coherence range at the matched K/d_{state} window.

coherence 0.2842 at construction, drifting to a final-checkpoint range-of- K -means of 0.373–0.385 under normal training); the $d = 128$ table (107 entities in 128 dimensions) is exactly orthogonal at construction ($\max |\cos| = 0.000000$). This is the single most natural remaining mechanistic candidate for the cliff's disappearance.

- We test it directly: holding $d_{\text{state}} = 128$, $n = 107$, $K = 68$ fixed at exactly the geometry Act 2 measured flat, we inject controlled, *frozen* coherence into the anchor table under a concentrated (rank-4) structure and sweep the dose — 0.130, 0.284, 0.40 — the last exceeding

$d = 64$'s own realized training band (Fig. 2).

- **Result:** $h4 = 1.0$ at every cell, every dose, 10/10, no exception — flat across the entire dose range, including 0.40. This is the strongest possible exoneration this design can produce for the rank-4 structure: directly-injected anchor-table coherence, held frozen for the full run and verified never to drift (bit-identical across all 10 recorded checkpoints per cell, zero slack), does not reproduce the $d = 64$ cliff at matched $K/d_{\text{state}}/n$ geometry.
- Frozen-constancy and instrument-saturation are both independently verified: `max_abs_cos` is exactly bit-identical across every checkpoint at every dose (the frozen-table mechanism worked as built), and hop-21 recovery ranges 0.9987–1.0 across all 10 cells (not uniformly 1.0) — the flat $h4 = 1.0$ is a real ceiling for this quantity, not a broken or saturated readout.

4 The Honest Open Question

- Neither of the two most naturally testable single-variable accounts of the cliff's dissolution survives: the K/d_{state} ratio (Act 2) is ruled out directly by the flat $d = 128$ result, and scalar, rank-4-structured anchor-table coherence (Act 3) is ruled out directly by the flat dose-response result — injecting coherence up to and beyond $d = 64$'s own realized training band still does not reproduce the cliff at $d = 128$.
- **Two candidates survive the waves above; a follow-up wave (below) has since tested one of them further.**
 1. **Overlap structure (diffuse vs. concentrated).** A concentrated (rank-4) injection collapses the table's n rows toward a shared low-dimensional subspace; a *diffuse* (`subspace_rank=48`) injection spreads the same scalar dose across a far less structured set of directions. It remains possible the cliff tracks *how* coherence is distributed across the table, not merely its scalar magnitude. This is a co-primary, pre-registered follow-on arm — design complete, not yet run.
 2. **Absolute state capacity.** d_{state} grew $4\times$ ($64 \rightarrow 128$) between the cliff's location and its disappearance, while K only grew $\sim 2\times$ (34–46 at $d = 64$ vs. 68–92 at $d = 128$, holding K/d_{state} fixed). It remains possible the relevant quantity is raw available state dimensionality itself (or slack $d_{\text{state}} - K$), not any ratio or coherence measure — a hypothesis no wave here can speak to, since d_{state} was never varied independently of K/d_{state} or coherence.
- **Candidate 2 has since been tested directly, and the ratio-invariance null it would need is REFUTED**

— **load tolerance grows SUPER-LINEARLY in d_{state} , not at a fixed K/d_{state} ratio.** A follow-up wave re-ran the identical localization fit at two further scales. Against a pre-registered ratio-invariant band $[0.4745, 0.6165]$ (centered on $d=64$'s own $x_0=0.5455$), $d_{\text{state}}=80$ locates its own transition at $x_0=0.6779$ (95% CI $[0.6683, 0.6867]$, a clean, non-degenerate fit, $n=5$ seeds/ K) — entirely above the band, a real exclusion of fixed-ratio invariance. **At $d_{\text{state}}=96$, a wide-grid attempt ($K=72, 78, 84, 90$) initially flagged 11 of 12 new cells inadmissible; an independently-audited repro instrument diagnosed a tolerance miscalibration in the convergence gate — all 4,608 flagged episodes resolve by $n_{\text{iter}} \leq 28$ against a production $n_{\text{iter}}=20$ — and only the admission flag, never the $h4$ measurement, was recalibrated, unlocking the full grid at zero new GPU spend.** The re-fit on the complete grid ($K \in \{69, 72, 78, 84, 90\}$, $n=3/K$) finds NO cliff anywhere through $K/d_{\text{state}}=0.9375$, but a non-monotonic curve ($h4$ means $0.9592/0.9216/0.9326/0.9581/1.0000$: a dip at $K=72/78$, partial recovery at $K=84$, an exact ceiling at $K=90$ in these archived seeds — §5 reports a fresh-seed follow-up that does not replicate this exact ceiling) that the sigmoid fit cannot localize — 100% of 4,000 bootstrap resamples land at the fit's own degenerate parameter bounds, from genuine per-seed scatter (the noisiest group, $K=72$, spans 0.8426 – 0.9904) rather than a data gap. An absolute-slack account ($x_0 \propto d_{\text{state}}$ minus a constant, band $[0.718, 0.739]$) and a power-law account (band $[0.768, 0.837]$) both remain live and neither is excluded — **the precise functional form of the super-linear growth, and the exact $d_{\text{state}}=96$ transition point, remain open.** Seed escalation at $K=72$ (the noisiest group) is the registered next step, not launched by this wave.

- Adjudicating candidate 1 still requires a not-yet-designed wave: a diffuse-structure Stage 2 (co-primary with the rank-4 result already reported). Not registered in detail as of this writing; a natural, concrete next step, stated here rather than left implicit.
- We report this as a genuinely open question, not a hedge: a precisely-located transition, its clean dissolution with scale, and the ruling-out of the single most natural mechanistic account of that dissolution is, on its own, a complete and informative capacity story — even though it does not yet identify what *does* drive the transition.

5 Limitations

- **Scope.** All results use a single-head, single-layer DeltaNet, a 107-entity synthetic-grammar testbed, and $\leq$ sub-1M trainable parameters at $d_{\text{state}} \in \{64, 128\}$

only ($d_{\text{state}} = 32$ is a hard kernel-level crash in the underlying `fla` dependency, measured independently of this program, not a cost tradeoff). No claim is made about production-scale Gated-DeltaNet [2, 8], matrix-valued RNNs at scale [3], or natural language.

- **Two d -points in Act 2, one direction — since extended to four, still not enough to pin the functional form.** Act 2’s own dimension comparison uses exactly two scales (64, 128); this can determine that the cliff shifted or vanished but cannot, by two points alone, establish which of several d -linked confounds (state size, optimization dynamics through a larger Newton–Schulz block, or table coherence) is causally responsible — which is exactly why Act 3 tests the coherence candidate directly rather than inferring it. A follow-up wave (§4) has since added two further scales ($d_{\text{state}}=80, 96$) and refutes fixed-ratio invariance outright, but even after a wide-grid follow-up resolved an admission-gate tolerance miscalibration and unlocked a full 5- K grid at $d=96$, the points there (a non-monotonic curve, no cliff through $K/d_{\text{state}}=0.9375$, 100% bootstrap-degenerate) are still not enough to distinguish an absolute-slack account from a power-law account or to fix the exact $d_{\text{state}}=96$ transition point — a seed escalation at the noisiest measured group ($K=72$) is the registered next step, not yet run.
- **The $d=96$ near-ceiling regime is itself instrument-sensitive, at two independent levels, per a dedicated follow-up diagnostic.** Built to test whether $K=84$ ’s own dip below $K=90$ is a genuine non-monotonicity or an eval-time artifact of overlapping entity pools, a margin-restricted re-run of $K=84$ and $K=90$ at fresh, independent seeds (already-escalated $n_{\text{iter}}=28$) landed at DEGENERATE_CELL: neither cell was admissible, so neither of the diagnostic’s own adjudication triggers fired. Three findings are disclosed as descriptive-only: (a) $K=84$ now fails a DIFFERENT admission leg (value-salvage $0.093 < 0.10$, otherwise-clean convergence) than the original grid’s failure mode (convergence failing with value-salvage passing) — the failing leg swapped; (b) $K=90$ stays fallback-inadmissible even at $n_{\text{iter}}=28$, showing the admission frontier itself moves with K/d_{state} rather than being fixed; (c) at this fresh seed $K=90$ ’s own $h4$ measured 0.9725, not the archived exact 1.0000 — that specific ceiling reading did not replicate — while $K=84$ ’s own pool-restriction shift ($+0.0330$) is $\approx 12\times$ the measured noise floor (0.0028), consistent with a real pool-margin effect in direction even though the routing correctly refuses to promote an inadmissible cell to a finding. The NO-CLIFF-through- $K/d_{\text{state}}=0.9375$ result above stands (a claim about the absence of a monotonic decline, not about the exact value at $K=90$), but its fine structure near the ceiling is now known to be instrument-limited rather than settled; the escalation is registered for a

design-level follow-up, not yet run.

- **The dose-response result is rank-4-only.** A rank-4-only exoneration cannot distinguish “coherence-as-a-scalar does not drive the cliff” from “coherence-of-this-particular-concentrated-structural-kind does not drive the cliff, but a diffuse injection might” — the diffuse co-primary arm (§4) remains unresolved.
- **No mechanism claim beyond the disclosed comparison axis.** Every wave in this paper measures *where* a transition sits (in K/d_{state} , in dose) and whether it is present or absent in a measured window; none constitutes a controlled test of *why* the underlying write mechanism behaves this way at the level of individual gradient updates.
- **Learned-table reference point is non-dispositive.** A separate, non-frozen $d = 128$ zero-dose reference point (final coherence ≈ 0.13 – 0.20 , drifted from init under ordinary training, $h4 = 1.0$) is reported honestly as a low-dose null datum in the full paper but is not a substitute for the frozen, controlled dose sweep this paper’s Act 3 depends on — it confounds coherence magnitude with how that coherence was arrived at.
- **Single architecture family.** All results are specific to the delta-rule state-update; whether the same three-act structure (cliff, dissolution, coherence-exoneration) appears in other fast-weight or matrix-valued architectures [3, 6] is untested.

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